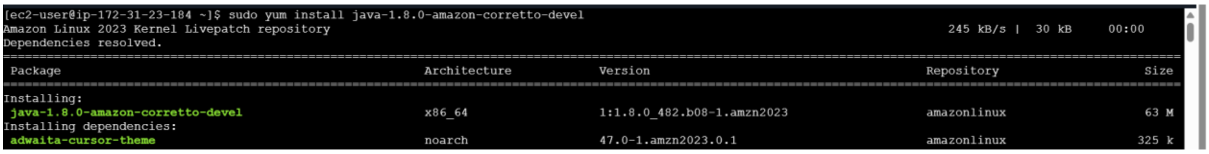
Project on Freestyle Jobs

To Begin with the Lab

Summary of the Lab

This lab demonstrates deploying a Java web application using **Jenkins Freestyle Jobs** with two EC2 instances: a **Jenkins server** and a **Tomcat server**. The Tomcat server is configured by installing **Java (Amazon Corretto 1.8)** and **Apache Tomcat**, and verifying it through port **8080**. On the Jenkins server, the **Publish Over SSH plugin** and **Maven** are configured. A Freestyle job is created that pulls source code from **GitHub**, builds the project using **Maven (compile and package)**, and generates a WAR file. Jenkins then transfers the WAR file to the Tomcat server using SSH, deploys it in the **webapps** directory, and the application is accessed via the Tomcat URL.

- EC2 Instance (Two servers) are used:
- Jenkins Server:
    - Maven, Publish Over SSH plugin
- Tomcat Server:
    - Java, Apache Tomcat
- Create a new EC2 instance.
- Recommended configuration:
- Instance type:
    - t2.medium
- Storage:
    - 15 GB
- Security Group:
    - Allow HTTP (80),
    - Allow HTTPS (443),
    - Allow 8080, Allow SSH (22)
- Launch the instance.
- Connect to the server using SSH.
- Install Java:
    - sudo yum install java-1.8.0-amazon-corretto-devel



- Verify:
- java -version

```
[ec2-user@ip-172-31-23-184 ~]$ java -version
openjdk version "1.8.0_482"
OpenJDK Runtime Environment Corretto-8.482.08.1 (build 1.8.0_482-b08)
OpenJDK 64-Bit Server VM Corretto-8.482.08.1 (build 25.482-b08, mixed mode)
[ec2-user@ip-172-31-23-184 ~]$
```

- Install Apache Tomcat
- [Download Tomcat:]()

Binary Distributions

- Core:
  - zip (pgp, sha512)
  - tar.gz (pgp, sha512)
  - 32-bit Windows zip (pgp, sha512)

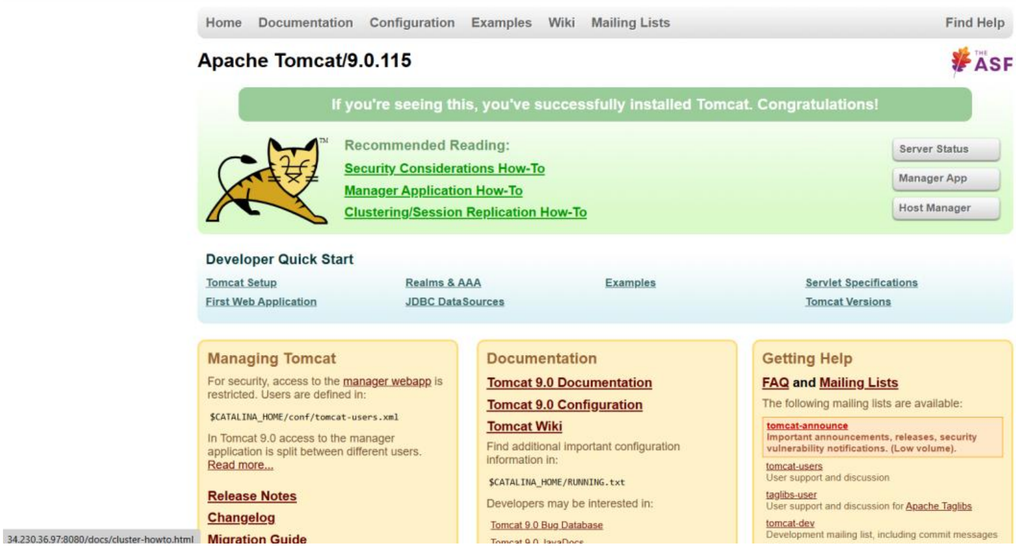
- wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.115/bin/apache-tomcat-9.0.115.tar.gz
- Extract the package:
  - tar xvzf apache-tomcat-9.0.115.tar.gz
- Move inside Tomcat directory:
  - cd cd apache-tomcat-9.0.115/bin/
- Look for the startup.sh file

```
[ec2-user@ip-172-31-23-184 ~]$ cd apache-tomcat-9.0.115/bin/
[ec2-user@ip-172-31-23-184 bin]$ ls
bootstrap.jar        catalina.sh   commons-daemon-native.tar.gz  configtest.sh  digest.sh     setclasspath.bat  shutdown.sh   tomcat-juli.jar         tool-wrapper.sh
catalina-tasks.xml   ciphers.bat   commons-daemon.jar            daemon.sh      makebase.bat  setclasspath.sh   startup.bat   tomcat-native.tar.gz    version.bat
catalina.bat         ciphers.sh    configtest.bat                digest.bat     makebase.sh   shutdown.bat      startup.sh    tool-wrapper.bat        version.sh
[ec2-user@ip-172-31-23-184 bin]$
```

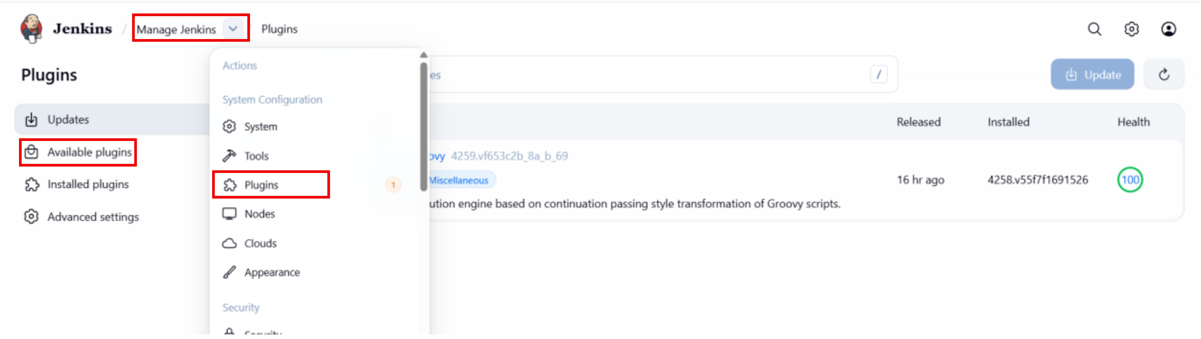
- Start Tomcat:
  - ./startup.sh

```
[ec2-user@ip-172-31-23-184 bin]$ ./startup.sh
Using CATALINA_BASE:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_HOME:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_TMPDIR: /home/ec2-user/apache-tomcat-9.0.115/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/ec2-user/apache-tomcat-9.0.115/bin/bootstrap.jar:/home/ec2-user/apache-tomcat-9.0.115/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[ec2-user@ip-172-31-23-184 bin]$
```

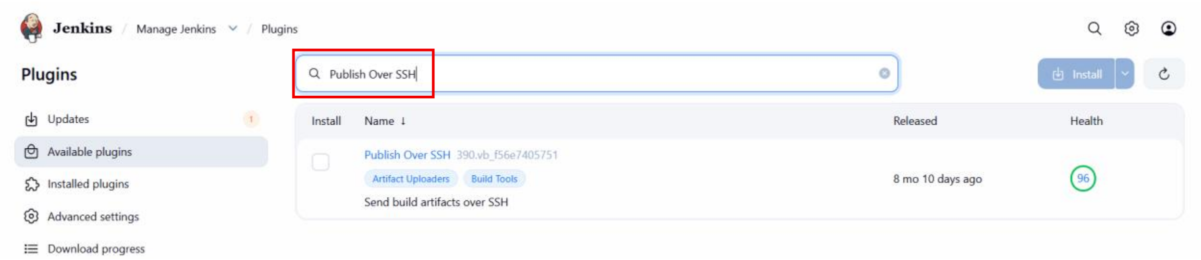
- Check Tomcat in browser:
  - http://<TOMCAT_PUBLIC_IP>:8080



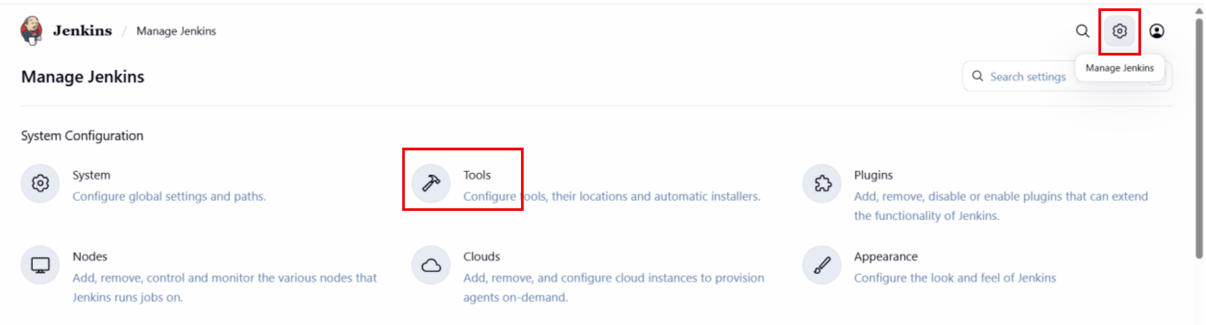
- If Tomcat page appears → installation successful.
- Stop Tomcat (for deployment):
    - ./shutdown.sh
- Install Jenkins Plugin (Publish Over SSH)
- Go to Jenkins Dashboard
- Manage Jenkins
- Plugins
- Available Plugins



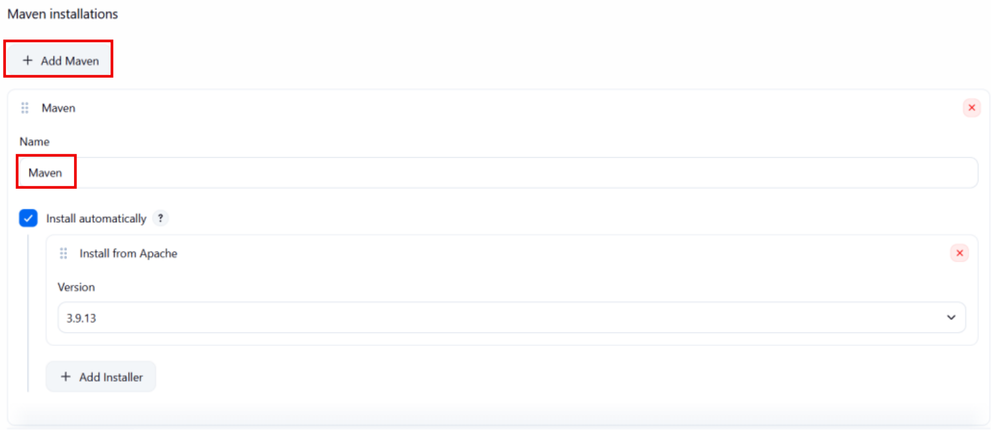
- Search: Publish Over SSH



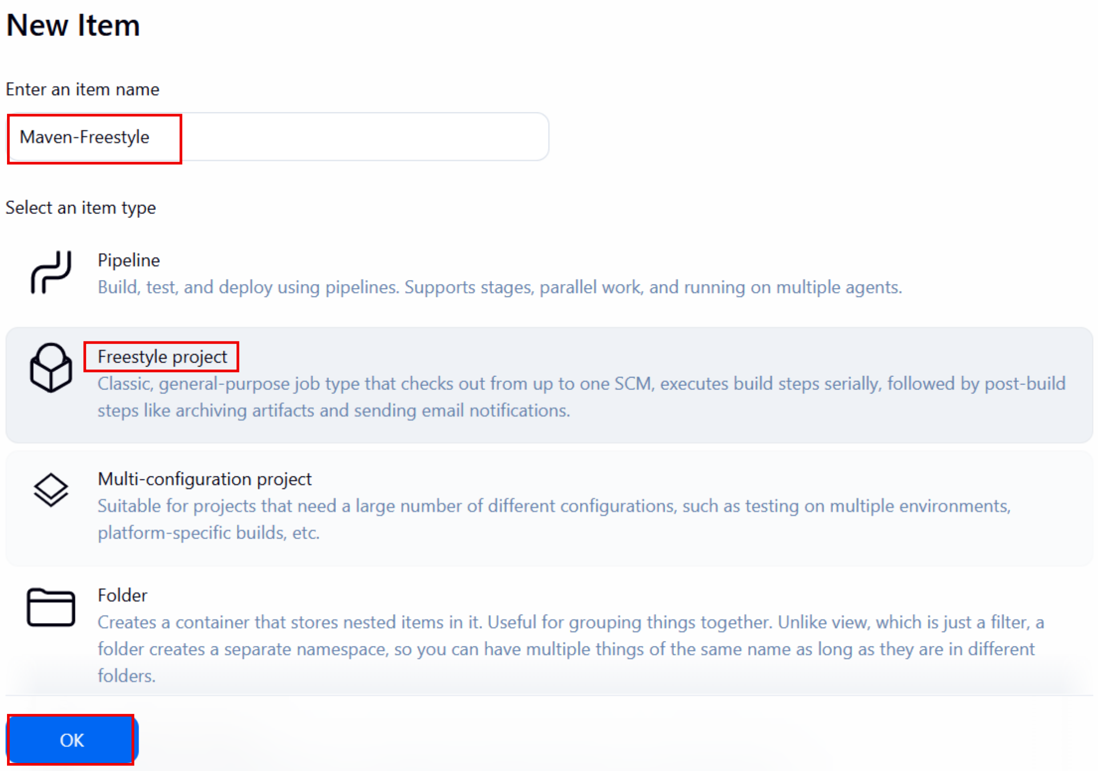
- Install the plugin.
- Open **Jenkins Dashboard**.
- Click **Manage Jenkins**.
- Click **Tool.**

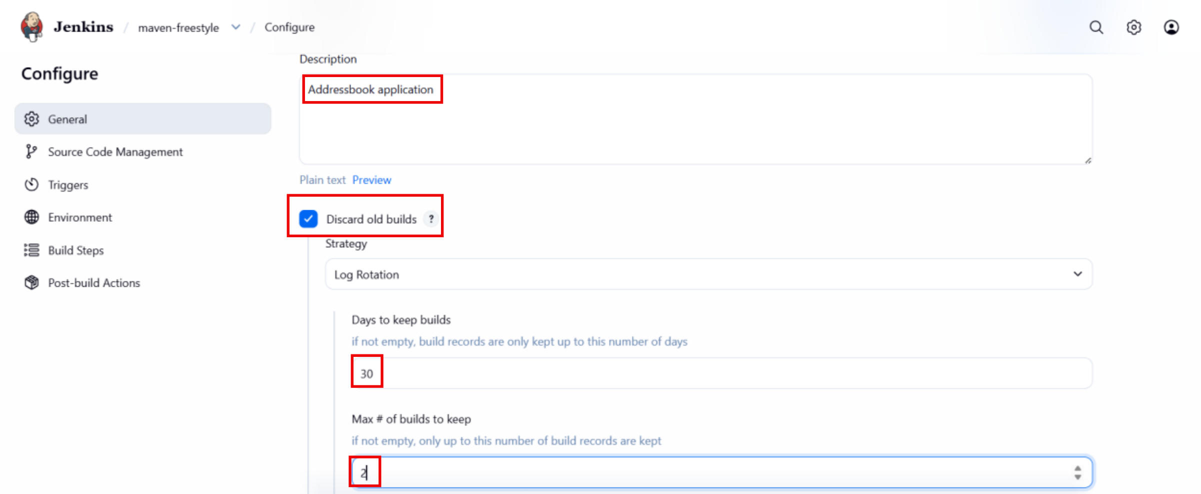


- Scroll to **Maven Installation**.
- Click **Add Maven**.

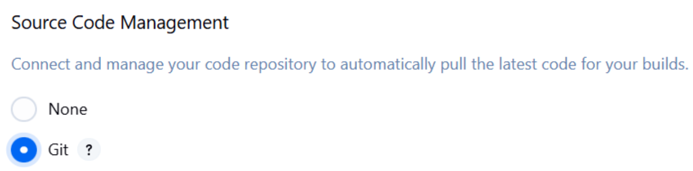


- Click Save.
- If you have already done with part you can skip.
- Go to Jenkins Dashboard.
- Click New Item.

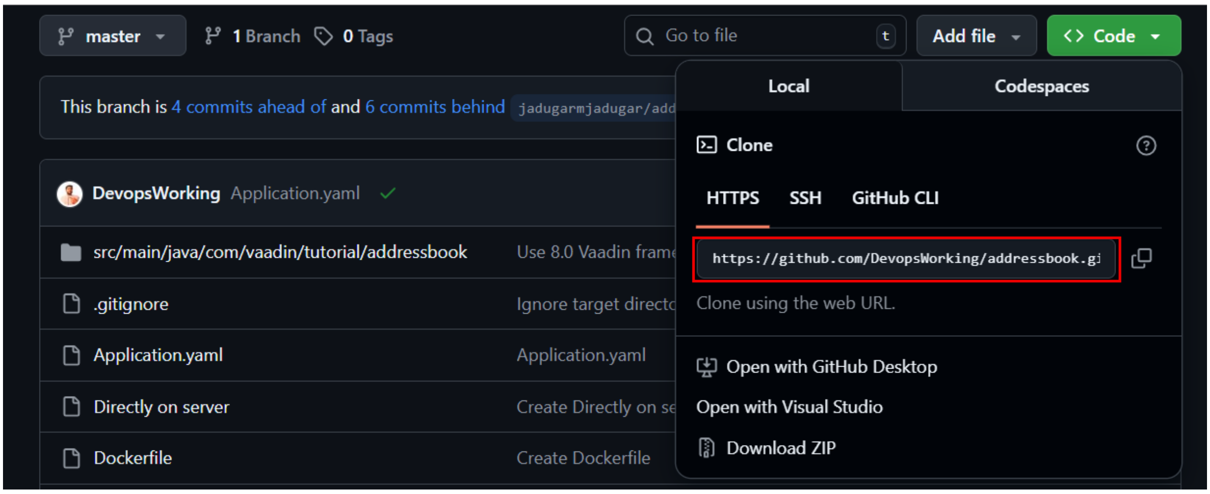




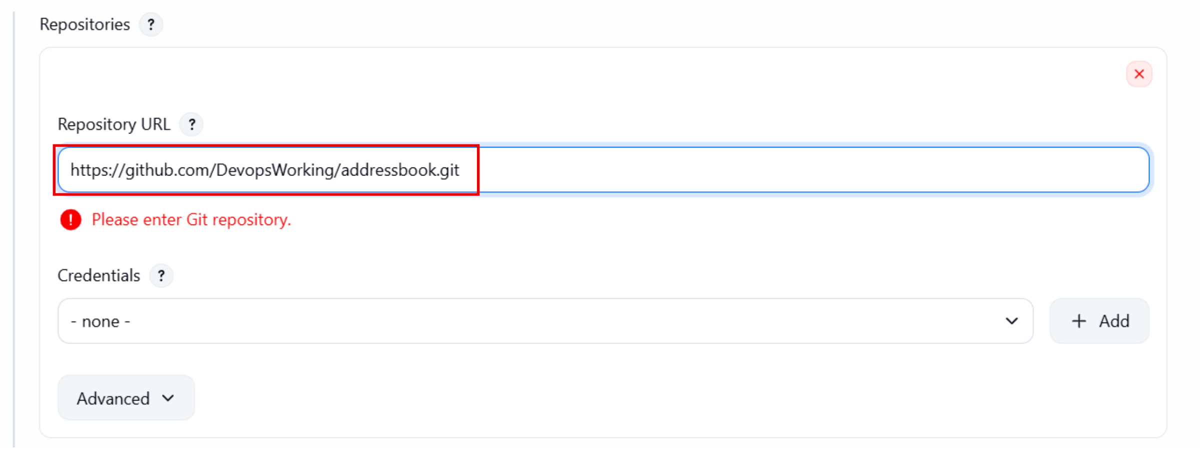
- Navigate Source Code Management
- Select the git option from there



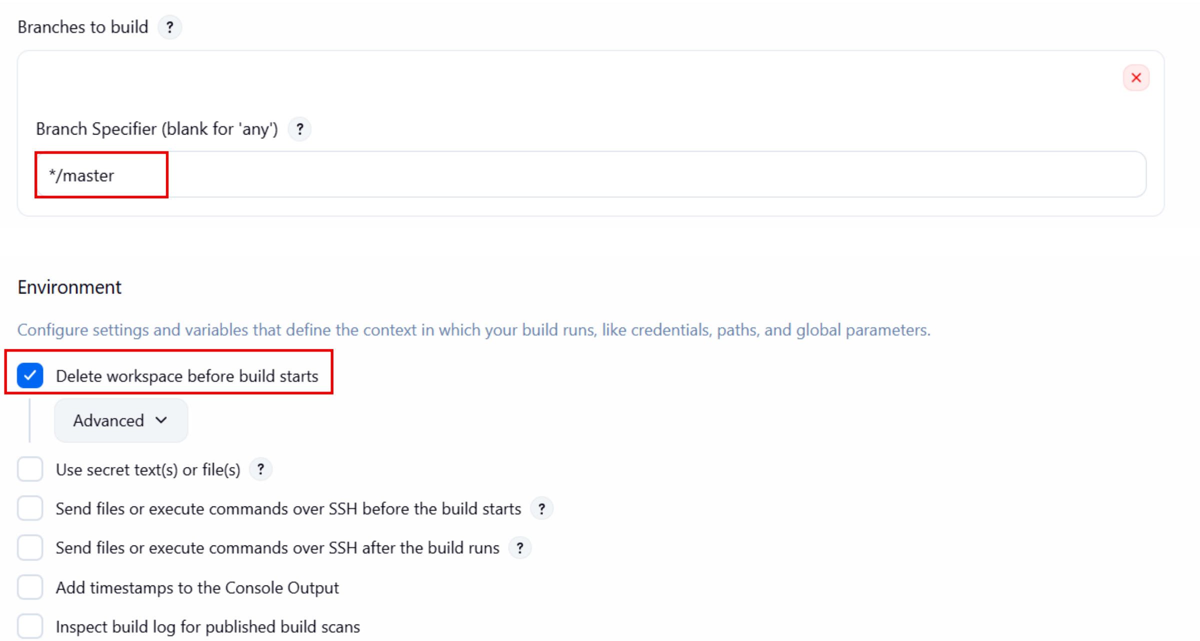
- After selecting git.
- Copy the URL from the github



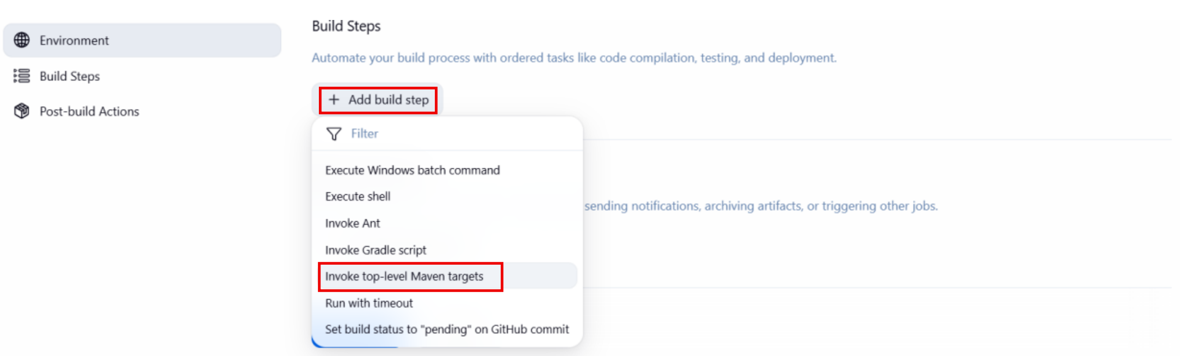
- Enter the Repository URL
- Note: If your repository is private, you have to select the credintials form the credential option.

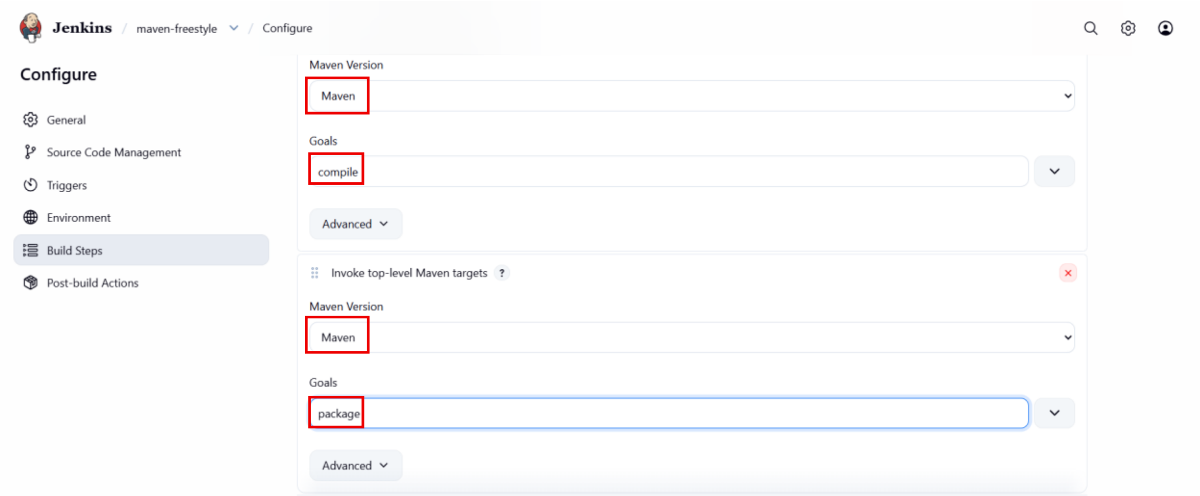


- Check you github repository is master or main.



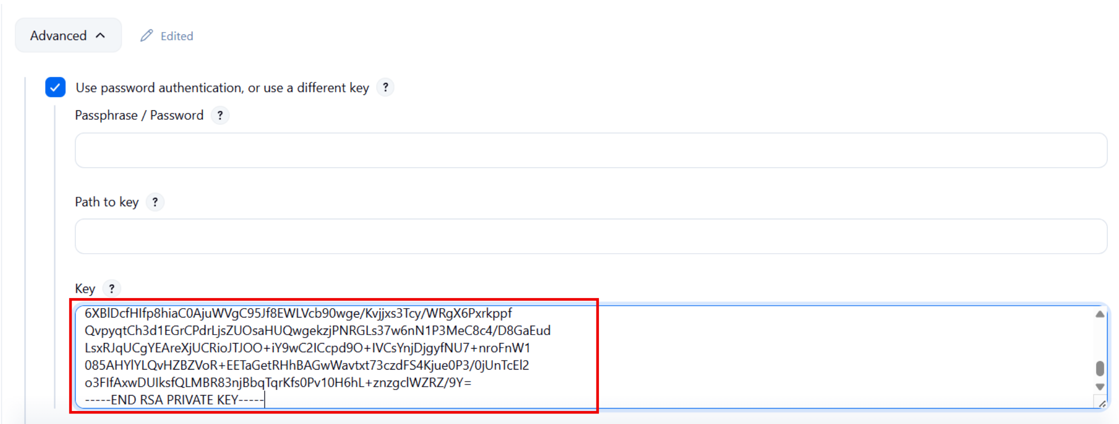
- Navigate to the Build Step section
- Click on the Add Build Step
- Select the 'Invoke top-level Maven targets'



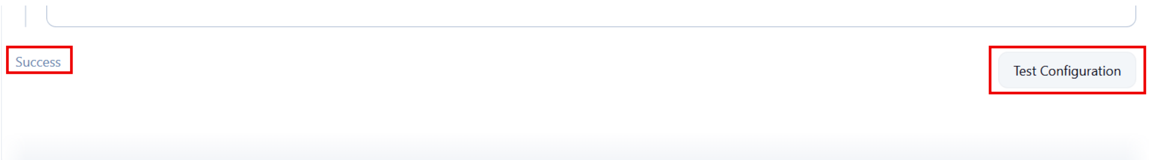


- Similarly make Goals for – compile and package
- Save the Job
- Now click on the Build now
- After saving it will create a job successfully
- Configure Publish Over SSH
- Go to:
    - Manage Jenkins
- Configure System
- Scroll to:
    - Publish Over SSH
- Click Add Server.
- Fill details:
- Name:
    - TomcatServer
- Hostname:
    - <Tomcat_Public_IP>
- Username:
    - ec2-user
- Remote Directory:
    - /home/ec2-user/apache-tomcat/webapps

- In Advanced section:
  - Use password authentication or use a different key
- Add Private Key
- Paste the EC2 private key (.pem file).

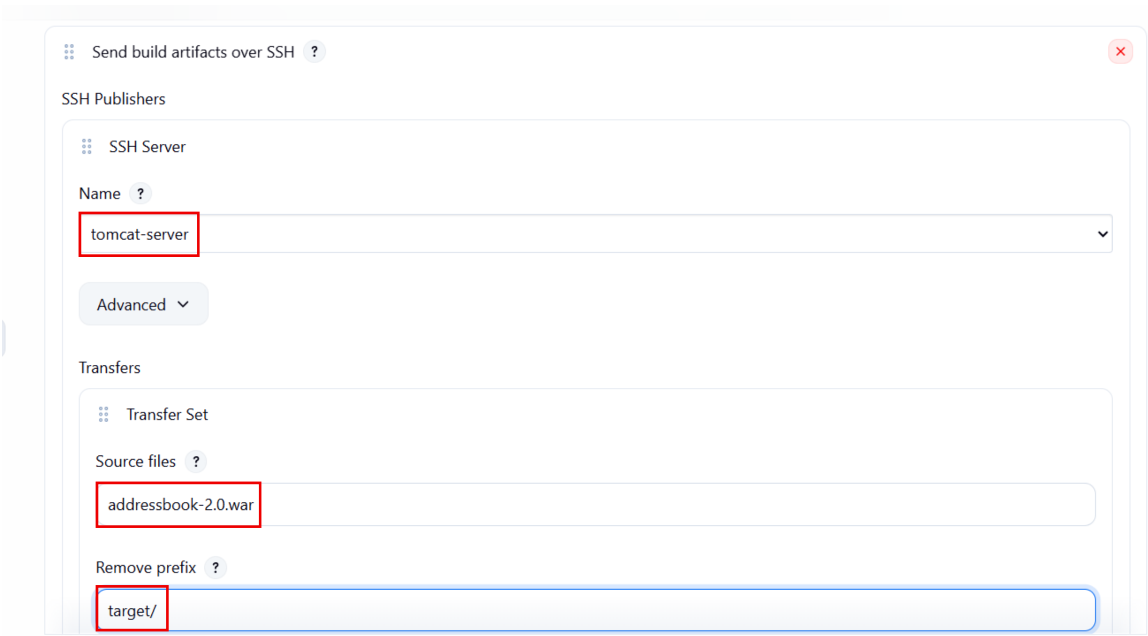


- Click:
  - Test Configuration
- If configuration is correct → Success message appears.



- Click Save.
- Open Jenkins job > Click Configure

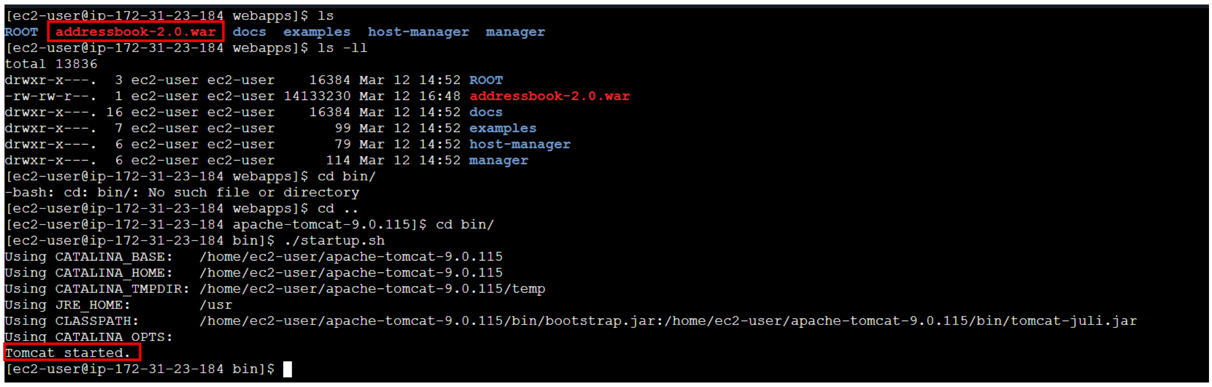
- Configure Artifact Transfer
- Look for **Post-build Actions**
- Select:
  - Send build artifacts over SSH
- Choose server:
  - Tomcat-Server
- Provide source file:
  - target/addressbook-2.0.war
- Remove prefix:
  - target/



- Click Save.
- Click on Build Now
- In console Output shows:
  - Transferred 1 file

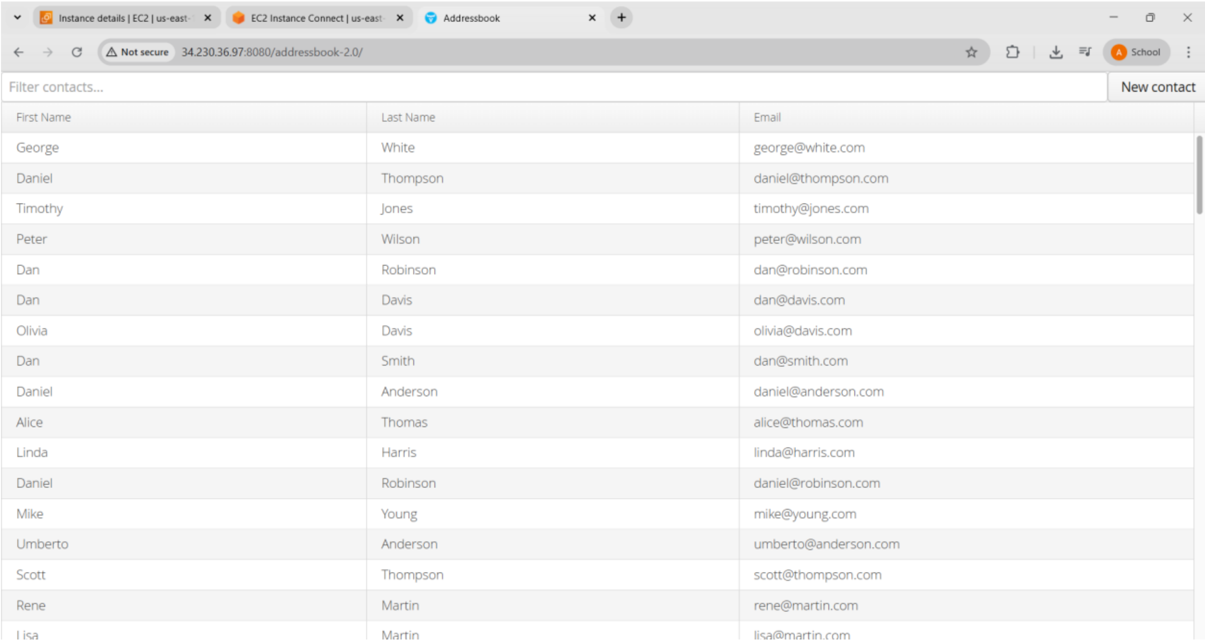
```
SSH: Connecting from host [ip-172-31-13-72.ec2.internal]
SSH: Connecting with configuration [tomcat-server] ...
SSH: Disconnecting configuration [tomcat-server] ...
SSH: Transferred 1 file(s)
Finished: SUCCESS
```

- Verify WAR File on Tomcat Server
- Login to Tomcat server:
  - ls
- You should see:
  - addressbook-2.0.war



```
[ec2-user@ip-172-31-23-184 webapps]$ ls
ROOT  addressbook-2.0.war  docs  examples  host-manager  manager
[ec2-user@ip-172-31-23-184 webapps]$ ls -ll
total 13836
drwxr-x---.  3 ec2-user ec2-user    16384 Mar 12 14:52 ROOT
-rw-rw-r--.  1 ec2-user ec2-user 14133230 Mar 12 16:48 addressbook-2.0.war
drwxr-x---. 16 ec2-user ec2-user    16384 Mar 12 14:52 docs
drwxr-x---.  7 ec2-user ec2-user       99 Mar 12 14:52 examples
drwxr-x---.  6 ec2-user ec2-user       79 Mar 12 14:52 host-manager
drwxr-x---.  6 ec2-user ec2-user      114 Mar 12 14:52 manager
[ec2-user@ip-172-31-23-184 webapps]$ cd bin/
-bash: cd: bin/: No such file or directory
[ec2-user@ip-172-31-23-184 webapps]$ cd ..
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ cd bin/
[ec2-user@ip-172-31-23-184 bin]$ ./startup.sh
Using CATALINA_BASE:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_HOME:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_TMPDIR: /home/ec2-user/apache-tomcat-9.0.115/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/ec2-user/apache-tomcat-9.0.115/bin/bootstrap.jar:/home/ec2-user/apache-tomcat-9.0.115/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[ec2-user@ip-172-31-23-184 bin]$
```

- Access Application
- Open browser:
  - http://<Tomcat_Public_IP>:8080/addressbook-2.0
- The Address Book Application UI should appear.



Final Workflow

GitHub Repo

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Jenkins Freestyle Job

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Maven Build (compile + package)

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WAR File created

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Publish Over SSH Plugin

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WAR copied to Tomcat server

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Tomcat deploys application